

Daniel Sim

Senior Software Engineer | C#, ASP.NET, Azure, AWS

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ABOUT ME

Senior software engineer with 11 years of experience, specializing in **C#, ASP.NET, Cloud Infrastructure** and **Microservices** architecture. Demonstrated capabilities in transforming business needs into software solutions and balancing rapid delivery with high quality standards in 3 big projects.

SKILLS

C#, ASP.NET, ASP.NET MVC, .NET Core, JS, TS, React, Angular, Blazor, AWS, AWS Lambda, Amazon EKS, Azure, Azure functions, Azure DevOps, AKS, MS SQL Server, PostgreSQL, MySQL, Amazon DynamoDB, Azure CosmosDB, Entity Framework, EF Core, ADO.NET, Dapper, Docker, Terraform, Git, Agile/Scrum, XUnit, NUnit, Microservices, SOAP, HIPAA, RabbitMQ, Datadog, Grafana

EXPERIENCE

CVS Health Woonsocket, RI

Remote

Lead Backend Engineer

Nov 2024 – Feb 2025

- Led the migration of legacy monolithic healthcare applications to **Azure cloud-native Microservices** architecture using **AKS** and **Apache Kafka**, optimizing performance and scalability while reducing operational costs by 30%.
- Re-architected **Azure Data Factory** and **ETL** pipelines for growth metrics and clinical quality data to capture change data which not only reduced spend by 50% but also 3x data freshness.
- Converted an **EMR** management board with **ASP.NET MVC** and **Blazor** into web application based on **Angular** and **ASP.NET Core**, focusing on **HL7** standards to ensure secure and inter-operable data exchange.
- Re-designed critical **API** endpoints with **gRPC**, optimizing database queries, reducing latency by 25% for high-demand healthcare workflows.
- Collaborated with **cross-functional** teams, including frontend developers and product managers, to improve application performance and user experience.

Forward San Francisco, CA

Remote

Staff Software Engineer

Jul 2021 – Oct 2024

- Developed a secure **EHR** system using **ASP.NET Core 7**, **RabbitMQ**, and **Angular 17**, incorporating patient demographics management, medication tracking, and clinical notes, while ensuring **HIPAA** compliance through data encryption, audit logging, and strict access controls to enhance patient information security.
- Built a fully compliant v4.0.1 **FHIR** Server with a Document Store database, and all CRUD operations for resource exposed through REST and **GraphQL**.
- Enabled performing labs in the **CarePod**, completely autonomous clinics, through APIs that orchestrated the collecting of blood, processing of the sample, sending the sample to external labs, and adding results to the **EMR** system for communication to members after reviews by the Clinical Team.
- Optimized data pipelines for healthcare analytics using **Databricks** notebooks with **PySpark** and **Scala**, improving processing efficiency by 30% and enabling real-time insights from **Azure Data Lake**.
- Implemented **LINQ** with **Lambda** on **SQL/NoSQL**, optimized performance using **Redis Cache**, and coded **LINQ** queries to customize data from **Azure Cosmos DB** in **Web API** actions.
- Focused on performance tuning by applying caching mechanisms such as **Redis** and utilizing **Grafana** and **Azure Monitor** for monitoring and logging.
- Enhanced backend API security by implementing **OAuth 2.0**, **JWT** authentication, IP whitelisting, and secure token storage, ensuring compliance with financial regulations and safeguarding sensitive data.
- Implemented automated **CI/CD** pipelines via **Azure Devops**, enabling microservices configuration without impacting live workflows, achieving 85% test coverage and reducing testing time by 50% through automated testing with **xUnit** and **SonarQube**.

Kroger Cincinnati, OH

Hybrid

Senior Software Engineer

Jul 2020 – Jun 2021

- Migrated a monolithic billing system to a microservices architecture using **ASP.NET Core**, **React** and **AWS Lambda**, implementing payment processing and invoicing while integrating **Microsoft Dynamic 365**.
- Built a **Store-and-Forward** system by adding an auxiliary bank server to temporarily store failed transactions and automatically reprocess after 1 min.
- Applied **secure SDLC** process, security review on new component architecture, using Fortify for **SAST** at Dev stage, using Acunetix for **DAST** and **WAF** at QA stage, **NoName** security and WAF for Prod stage.
- Integrated **Datadog APM** to trace microservices running on **Azure Container Apps**, identify and resolve latency issues, and configure custom **Datadog dashboards** to monitor performance metrics for **AWS Elastic Beanstalk**, enhancing system observability.
- Enabled interactive financial data visualization by integrating **Power BI** and **Tableau** dashboards, improving analysis of transaction trends, payroll summaries, and payment analytics, driving a 30% improvement in decision-making efficiency.

Redbox Chicago, IL

Onsite

Technical Lead

May 2013 – Feb 2020

- Conducted **code reviews** and mentored 3 junior developers, fostering a collaborative environment and strengthening a team of over 10 developers.
- Deployed a micro frontend system with **React**, enabling independent module deployment and enhancing scalability and optimizing API communication through **lazy loading**.

EDUCATION

UNIVERSITY OF ILLINOIS CHICAGO

Chicago, IL

Bachelor's degree, Mathematics

Aug 2010 – May 2015